



PS3: Estimation Windows and Portfolio Risk

CHEME 4/5660 Cornell University | Mathematical Companion

Disclaimer and Risks

Training only. This assignment uses historical prices and simplified investment assumptions for teaching. It provides no investment advice.

Short positions. Borrowed shares must be returned. A rising stock price can produce losses larger than the initial short-sale proceeds.

Exercise assumptions. Fractional shares are allowed. Short-sale proceeds finance long holdings. We model borrowing fees, while omitting dividends, other trading costs, collateral requirements, and forced liquidation.

The Portfolio Question

Does using more recent data lead to a better portfolio, and does allowing short positions help?

- **Choose the allocations.** Compare estimated risk with and without short positions, using 2014–2025 and then 2025 alone.
- **Test the holdings.** Evaluate their 2026 wealth and risk, then calculate what borrowing the shares costs.
- **Assess the evidence.** Interpret the observed comparison. Advanced adds modeled probabilities of beating the benchmark.

Reading guide Studies 1--3 are shared. The final MA-GBM study is Advanced only.

Study 1: Change the Estimation Window

Keep the eight stocks, purchase date, and optimization rules unchanged.

- **Long history:** all 3,017 prices from 2014–2025, giving 3,016 consecutive growth observations.
- **Recent history:** first select the 250 prices dated in 2025, then calculate 249 growth observations.

The recent window excludes the change from December 31, 2024 to January 2, 2025. The long window includes it. Both end at the same purchase prices.

Comparison Choose long-only and shorts-allowed GMV for each window. Equal weight supplies one historical reference.

Study 1: Estimate the Growth Rates

Within the selected window, let $S_{i,k}$ be stock i 's price in USD/share at observation k , for $i = 1, \dots, 8$. With $\Delta t = 1/252$ trading year, its growth rate is:

$$g_{i,k} = \frac{\log(S_{i,k}/S_{i,k-1})}{\Delta t}, \quad k = 1, \dots, n.$$

Use $n = 3016$ for 2014–2025 or $n = 249$ for 2025 alone. Write $\mathbf{g}_k$ for the growth vector. The sample mean and covariance are:

$$\hat{\mathbf{g}} = \frac{1}{n} \sum_{k=1}^n \mathbf{g}_k, \quad \hat{\Sigma}_g = \frac{1}{n-1} \sum_{k=1}^n (\mathbf{g}_k - \hat{\mathbf{g}})(\mathbf{g}_k - \hat{\mathbf{g}})^\top.$$

Units Growth is in inverse years; growth-rate covariance is in inverse years squared.

Study 1: Choose the Allocations

Let w_i be stock i 's fraction of initial wealth, and let $\mathbf{1}$ be the vector of eight ones. Both optimized portfolios solve:

$$\min_{\mathbf{w}} \mathbf{w}^\top \widehat{\Sigma}_g \mathbf{w}, \quad \mathbf{1}^\top \mathbf{w} = 1.$$

- **Long only:** also require $w_i \geq 0$ for every stock.
- **Shorts allowed:** permit negative weights, with no position bounds.
- **Equal weight:** use $w_i = 1/8$ as the reference allocation.

Estimated risk is $\hat{\sigma}_p = \sqrt{\mathbf{w}^\top \widehat{\Sigma}_g \mathbf{w}}$, in inverse years. It uses the lecture's linear growth approximation at the initial weights.

Scope No growth target or borrowing cost enters the optimization.

Study 1: Solve the Shorts-Allowed Problem

For a positive-definite sample covariance $\widehat{\Sigma}_g$ and the vector of ones $\mathbf{1}$, the global minimum-variance allocation is:

$$\widehat{\Sigma}_g \mathbf{x} = \mathbf{1}, \quad \mathbf{w}_{\text{shorts}} = \frac{\mathbf{x}}{\mathbf{1}^\top \mathbf{x}}.$$

Solve the linear system for $\mathbf{x}$; do not explicitly form the inverse. In Julia, use `covariance \ ones(8)`.

Every long-only allocation is feasible when shorts are allowed. The shorts-allowed minimum estimated variance therefore cannot be larger.

That conclusion concerns the **same estimated covariance**. It does not guarantee lower realized risk on the 2026 prices.

Study 1: Why Might a Volatile Stock Be Shorted?

For fixed initial weights, define the weighted growth of the long positions as G_+ and the growth of the shares owed as G_- :

$$G_+ = \sum_{i:w_i>0} w_i g_i, \quad G_- = \sum_{i:w_i<0} |w_i| g_i.$$

The portfolio's linear growth approximation is $G_+ - G_-$. Its variance is:

$$\text{Var}(G_+ - G_-) = \text{Var}(G_+) + \text{Var}(G_-) - \underbrace{2 \text{Cov}(G_+, G_-)}_{\text{offsetting movement}}.$$

Positive covariance provides an offset, while the short positions also contribute their own variance. **Individual volatility alone does not establish whether a short is a useful hedge.**

Study 1: Interpret the Financing

For weights w_i summing to one, define long exposure L , short exposure H , and gross exposure E , all relative to initial wealth:

$$L = \sum_i \max(w_i, 0), \quad H = \sum_i \max(-w_i, 0), \quad E = L + H.$$

The budget constraint gives $L - H = 1$. A short finances additional long holdings.

For example, with initial wealth of USD 10,000, $L = 1.20$ and $H = 0.20$ mean:

- USD 12,000 of long holdings;
- shares owed initially worth USD 2,000;
- USD 14,000 of gross exposure and USD 10,000 of net wealth.

Study 2: Hold the Shares

Let $W_0 = 10,000$ USD and $S_{i,0}$ be the December 31, 2025 price. The signed share count q_i and wealth before fees are:

$$q_i = \frac{W_0 w_i}{S_{i,0}}, \quad W_k^{\text{gross}} = \sum_i q_i S_{i,k}.$$

A negative q_i represents shares owed. **Share counts remain fixed** through day $N = 126$. Weights change with prices.

Day 0 is the final 2025 observation. Day 1 is the first 2026 observation. An N -day evaluation uses $N + 1$ price rows, including day 0.

Terminology Gross wealth means wealth before borrowing fees. Gross exposure measures the magnitude of all positions.

Study 2: Measure Realized Risk

For positive gross wealth at every observation, define the portfolio's realized growth $g_{p,k}$ and its average $\bar{g}_p$ over N intervals:

$$g_{p,k} = \frac{\log(W_k^{\text{gross}}/W_{k-1}^{\text{gross}})}{\Delta t}, \quad \bar{g}_p = \frac{1}{N} \sum_{k=1}^N g_{p,k}.$$

The realized growth-rate standard deviation is:

$$\sigma_{p,\text{realized}} = \sqrt{\frac{1}{N-1} \sum_{k=1}^N (g_{p,k} - \bar{g}_p)^2}.$$

Report it in inverse years, using wealth **before fees**. If gross wealth is ever nonpositive, this log-growth measure is unavailable.

Study 2: Separate Profit from Hedging

For signed share count q_i and sale day N , stock i 's dollar gain or loss, Π_i , is:

$$\Pi_i = q_i(S_{i,N} - S_{i,0}), \quad \Pi_{\text{short}} = \sum_{i:q_i < 0} \Pi_i.$$

A short gains when its price falls and loses when its price rises.

Compare **terminal wealth** and **realized risk** separately. A short can lose money while offsetting fluctuations in the long holdings.

The two optimized portfolios also have different long positions. Their performance difference cannot be attributed entirely to the shorts.

Study 2: Accumulate Borrowing Fees

Let B_k be the market value of the shares owed at the start of interval k , in USD. For annual borrowing rate b , total fees through day N are:

$$B_k = \sum_{i:q_i < 0} |q_i| S_{i,k}, \quad C_N(b) = b\Delta t \sum_{k=0}^{N-1} B_k.$$

Use $b = 0$ and $b = 0.03$ with the same holdings and price path.

- Charge on beginning-of-interval values; exclude the sale-day value.
- Accrue fees without interest and pay them at liquidation.
- Portfolios without short positions owe zero fees.

Convention These are illustrative annual rates using 252 trading days per year.

Study 2: Net Wealth and Benchmark Success

Net terminal wealth is gross wealth less borrowing fees. For benchmark rate $g_y = 0.05$ per trading year and holding time $T = N\Delta t$, define:

$$W_N^{\text{net}} = W_N^{\text{gross}} - C_N(b), \quad W_N^{\text{benchmark}} = W_0 e^{g_y T}.$$

Scaled net present value ρ_N discounts the proceeds to day 0 and compares them with the initial investment:

$$\rho_N = \underbrace{\frac{W_N^{\text{net}}}{W_0}}_{\text{wealth ratio}} \underbrace{e^{-g_y T}}_{\text{discount factor}} - 1.$$

Success means $\rho_N > 0$. Equality does not count. Increasing the fee reduces net wealth while leaving the holdings and risk before fees unchanged.

Study 3: Interpret the Window Comparison

Each window estimates its own covariance. Its GMV allocation minimizes variance under **that estimate**. Mean growth does not affect these weights because there is no growth target.

Compare all four allocations on the same 126-day 2026 price history. Use realized risk and net wealth to assess what carried over.

Changing the window changes both sample size and recency. One shared market episode cannot establish which window or shorting rule will always work best.

Advanced: Construct the MA-GBM Model

Fit one model to each window. For its mean growth $\hat{g}$ and covariance $\hat{\Sigma}_g$, define diffusion covariance D , its lower Cholesky factor A , and price drift μ_i by:

$$D = \Delta t \hat{\Sigma}_g, \quad AA^\top = D, \quad \mu_i = \hat{g}_i + \frac{1}{2} D_{ii}.$$

With independent standard Brownian motions $W_j(t)$, stock i follows:

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sum_{j=1}^8 A_{ij} dW_j(t).$$

D has units of inverse years; A has units of inverse square-root years. Keep the benchmark out of the fitted price drifts.

Advanced: Simulate Joint Price Paths

At interval k , draw an eight-component standard normal vector $\mathbf{z}_k$ with independent components. Its mean is zero and covariance is the identity matrix. The exact daily MA-GBM update is:

$$\log\left(\frac{\mathbf{S}_{k+1}}{\mathbf{S}_k}\right) = \hat{\mathbf{g}}\Delta t + A\sqrt{\Delta t}\mathbf{z}_k.$$

Ratios and logarithms are componentwise. Use independent draws across days and paths. The matrix A introduces the fitted cross-stock dependence.

Generate 20,000 paths through day 126 per model. Within each model, share price paths across portfolios and fees. Across models, reuse the normal draws; each fitted model transforms them into different price paths.

Interpretation These forecasts use historical fitted drifts. They are not risk-neutral prices.

Advanced: Estimate the Success Probability

Let s index the $K = 20,000$ simulated paths. On each path, calculate net wealth using the fixed shares and that path's borrowing fees. Then:

$$\hat{p}_N(b) = \frac{1}{K} \sum_{s=1}^K \mathbf{1} \left[W_{N,s}^{\text{net}}(b) > W_0 e^{g_y N \Delta t} \right].$$

The indicator is one for success and zero otherwise. Keep every path in the denominator, including nonpositive wealth outcomes.

A signed sum of lognormal asset prices is generally not lognormal. Compute wealth from the joint asset paths. Under our no-forced-liquidation assumption, retain the holdings through the sale day even if wealth crosses zero earlier.

Advanced: Interpret the Probability Estimate

For estimated success probability $\hat{p}_N$ based on K independent paths, the Monte Carlo standard error is:

$$\text{SE}(\hat{p}_N) = \sqrt{\frac{\hat{p}_N(1 - \hat{p}_N)}{K}}.$$

Multiply by 100 to report it in percentage points.

A small standard error means the simulation estimates the **fitted model's probability** precisely. It does not establish that the model or its estimated parameters describe future prices correctly.

Using common paths, a higher borrowing rate cannot raise the estimated success probability. It may leave the estimate unchanged if no outcome crosses the benchmark threshold.

Summary

The studies compare two estimation windows on one common 2026 episode, with and without short positions and borrowing fees.

Key takeaways

- Covariance determines whether short positions help offset portfolio fluctuations; individual volatility alone is insufficient.
- Lower estimated variance does not guarantee lower realized risk or more terminal wealth. Borrowing fees reduce the proceeds.
- MA-GBM supplies a model-based success probability. Simulation precision and forecast accuracy are different questions.

Use these distinctions to answer the central portfolio question in PS3.